From the Strain Equation to the Compatibility Equation

The idea: eliminate the displacement

The strain equation gives 6 components ε_{ij} from only 3 unknowns u_i. Differentiating enough times lets us combine the ε's so that every u term cancels. What remains is a relation among the ε's alone: the compatibility equation. Throughout, a comma denotes partial differentiation, repeated indices are summed, and $\boldsymbol{u}$ is assumed of class C^3 so that derivatives commute (Schwarz's theorem).

1 The 2D case

In 2D, three strain components depend on two displacements u and v:

$$\varepsilon_{xx} = \frac{\partial u}{\partial x}, \qquad \varepsilon_{yy} = \frac{\partial v}{\partial y}, \qquad 2\varepsilon_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}.$$

Differentiate each equation so that every term contains the same third-order derivatives of u and v:

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} = \frac{\partial^3 u}{\partial x\, \partial y^2}, \qquad \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = \frac{\partial^3 v}{\partial y\, \partial x^2}, \qquad 2\frac{\partial^2 \varepsilon_{xy}}{\partial x\, \partial y} = \frac{\partial^3 u}{\partial x\, \partial y^2} + \frac{\partial^3 v}{\partial x^2\, \partial y}.$$

The right-hand side of the third equation is the sum of the first two, hence

$$\boxed{\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = 2\frac{\partial^2 \varepsilon_{xy}}{\partial x\, \partial y}.}$$

2 The general 3D case

Step 1 – differentiate the strain twice.

$$\varepsilon_{ij} = \tfrac{1}{2}\left(u_{i,j} + u_{j,i}\right) \quad \Longrightarrow \quad \varepsilon_{ij,kl} = \tfrac{1}{2}\left(u_{i,jkl} + u_{j,ikl}\right).$$

Step 2 – express the four terms of the compatibility equation.

Term	Value in terms of $\boldsymbol{u}$
$\varepsilon_{ij,kk}$	$\frac{1}{2}u_{i,jkk} + \frac{1}{2}u_{j,ikk}$
$\varepsilon_{kk,ij}$	$u_{k,kij}$
$\varepsilon_{jk,ik}$	$\frac{1}{2}u_{j,kik} + \frac{1}{2}u_{k,jik}$
$\varepsilon_{ik,jk}$	$\frac{1}{2}u_{i,kjk} + \frac{1}{2}u_{k,ijk}$

Step 3 – add the first two, subtract the last two. Since derivative order is irrelevant ($u_{j,kik} = u_{j,ikk}$, $u_{k,jik} = u_{k,kij}$, ...), the terms cancel in pairs:

- $\frac{1}{2}u_{i,jkk}$ cancels with $\frac{1}{2}u_{i,kjk}$;
- $\frac{1}{2}u_{j,ikk}$ cancels with $\frac{1}{2}u_{j,kik}$;
- $u_{k,kij}$ cancels with $\frac{1}{2}u_{k,jik} + \frac{1}{2}u_{k,ijk}$.

Therefore

$$\boxed{\frac{\partial^2 \varepsilon_{ij}}{\partial x_k \partial x_k} + \frac{\partial^2 \varepsilon_{kk}}{\partial x_i \partial x_j} - \frac{\partial^2 \varepsilon_{jk}}{\partial x_i \partial x_k} - \frac{\partial^2 \varepsilon_{ik}}{\partial x_j \partial x_k} = 0.}$$

3 A more systematic route: via the rotation

This route explains *where* the combination comes from, instead of checking it after the fact. Split the displacement gradient into its symmetric and antisymmetric parts:

$$u_{i,j} = \varepsilon_{ij} + \omega_{ij}, \qquad \omega_{ij} = \tfrac{1}{2}\left(u_{i,j} - u_{j,i}\right),$$

where ω is the infinitesimal rotation. One differentiation gives its gradient from $\underset{\sim}{\varepsilon}$ alone:

$$\omega_{ij,k} = \tfrac{1}{2}\left(u_{i,jk} - u_{j,ik}\right) = \varepsilon_{ik,j} - \varepsilon_{jk,i},$$

since $\varepsilon_{ik,j} - \varepsilon_{jk,i} = \frac{1}{2}\left(u_{i,kj} + u_{k,ij} - u_{j,ki} - u_{k,ji}\right) = \frac{1}{2}\left(u_{i,jk} - u_{j,ik}\right)$.

For ω to be a genuine function, its second derivatives must commute: $\omega_{ij,kl} = \omega_{ij,lk}$. Now

$$\omega_{ij,kl} = \varepsilon_{ik,jl} - \varepsilon_{jk,il}, \qquad \omega_{ij,lk} = \varepsilon_{il,jk} - \varepsilon_{jl,ik},$$

so the condition reads

$$\varepsilon_{ik,jl} + \varepsilon_{jl,ik} - \varepsilon_{il,jk} - \varepsilon_{jk,il} = 0, \tag{1}$$

which is the full Saint-Venant system (81 scalar equations, 6 of them independent in 3D).

Recovering the contracted form. Simply setting $l = k$ in (1) gives the trivial identity $0 = 0$. Instead, take the component with indices (i, k, j, k), that is $j \to k$, $k \to j$, $l \to k$, and sum over k:

$$\varepsilon_{ij,kk} + \varepsilon_{kk,ij} - \varepsilon_{ik,jk} - \varepsilon_{jk,ik} = 0,$$

which is the boxed equation of Section 2. In tensor notation the same condition reads

$$\mathrm{Inc}(\underset{\sim}{\varepsilon}) = \mathrm{rot}\,\mathrm{rot}(\underset{\sim}{\varepsilon}) = 0.$$

Why the condition is also sufficient. If (1) holds on a simply connected domain, the relation $\omega_{ij,k} = \varepsilon_{ik,j} - \varepsilon_{jk,i}$ can be integrated to obtain ω; then $u_{i,j} = \varepsilon_{ij} + \omega_{ij}$ can be integrated to obtain $\boldsymbol{u}$. The resulting displacement reproduces $\underset{\sim}{\varepsilon}$, and is unique up to a rigid-body motion.